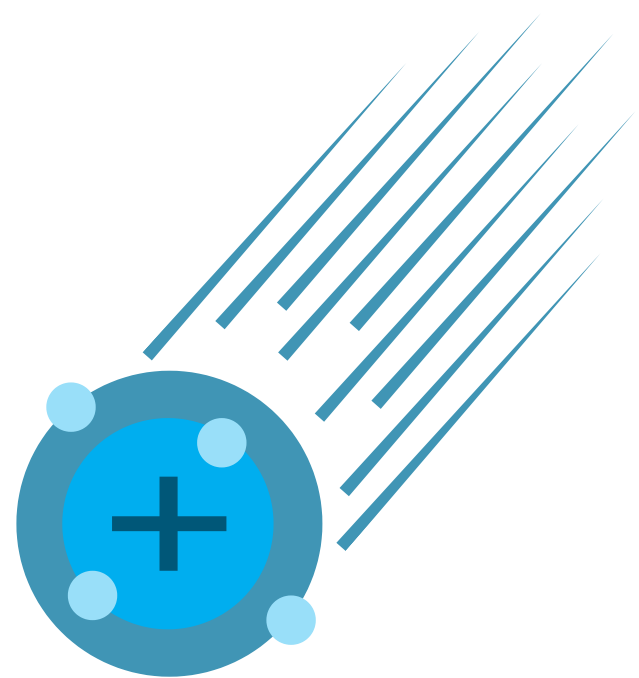
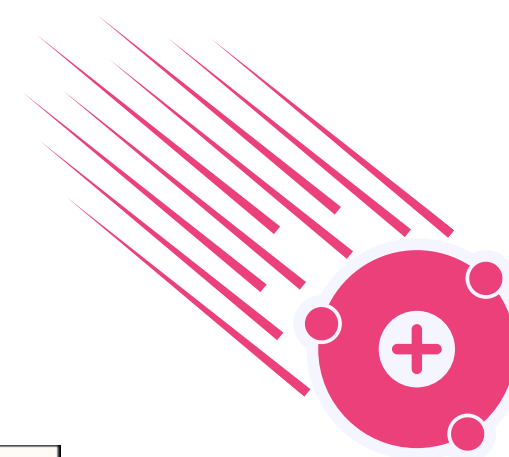




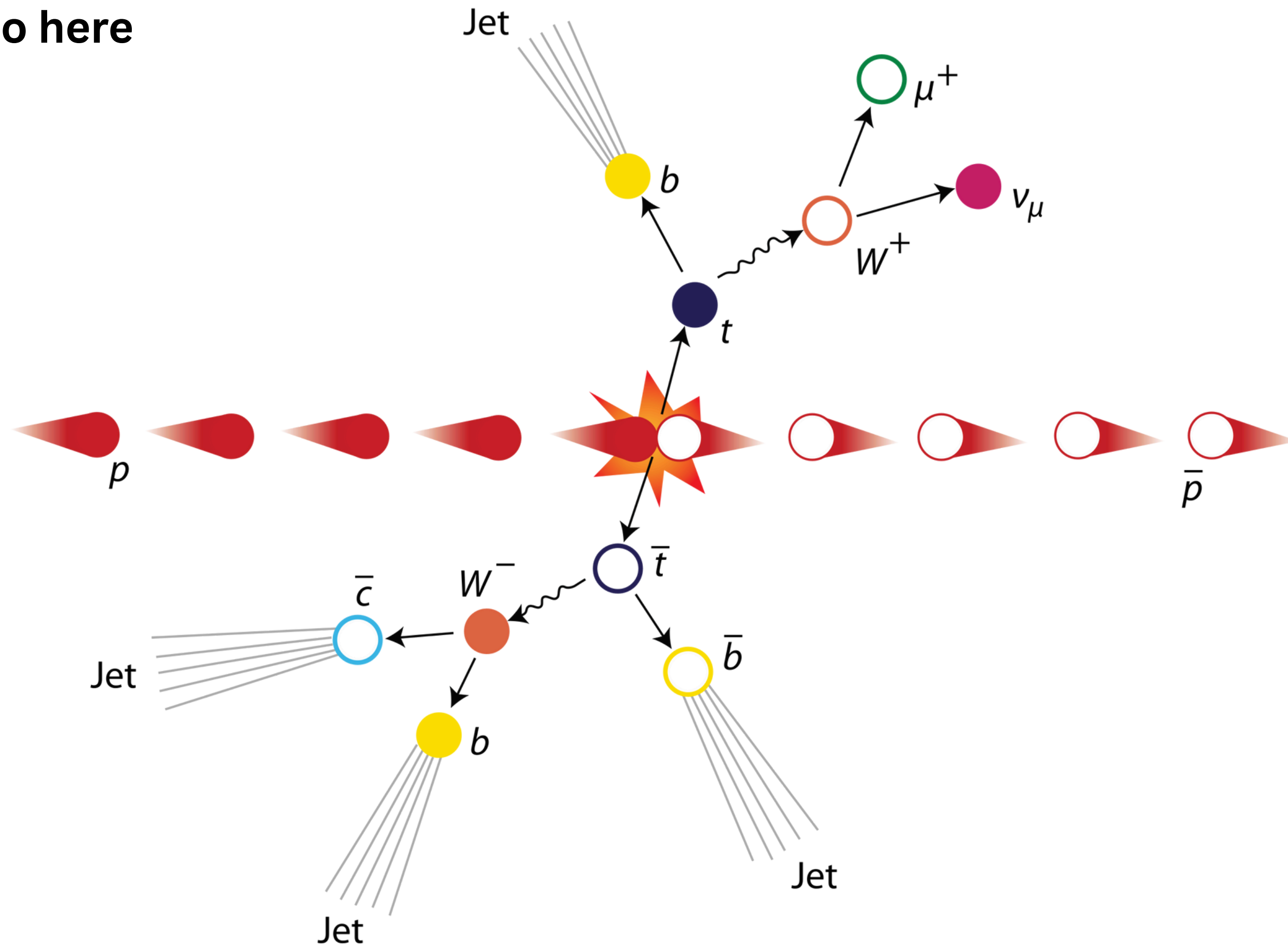
Particle Identification at CERN

Oleh Prostakov
Mykhailo Bondarenko
Anastasiia Mazur

Mentor: Pavlo Kashko

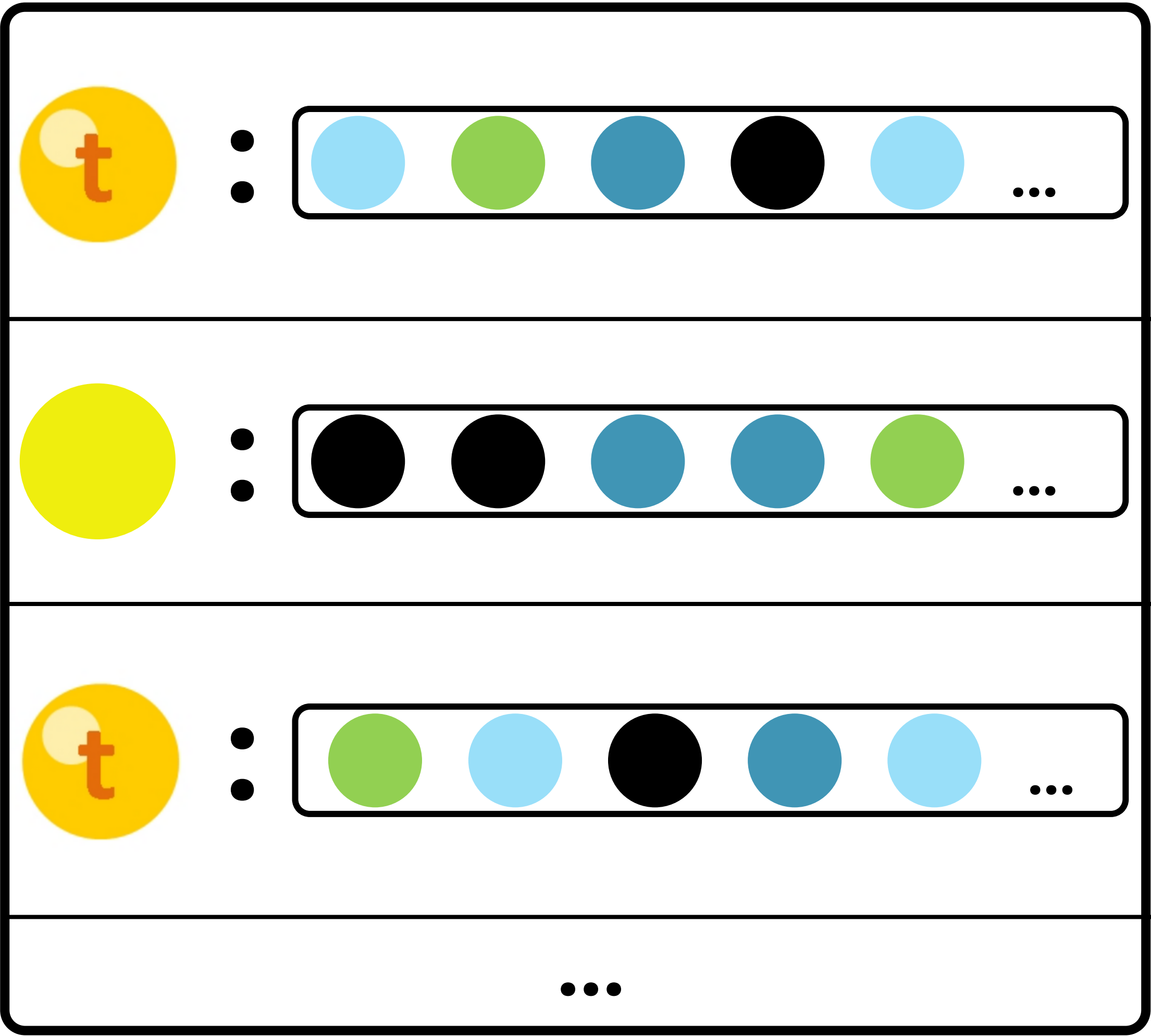


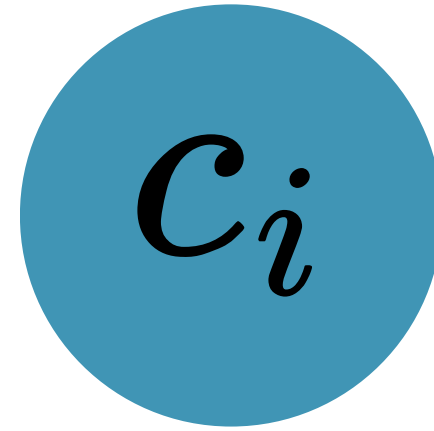
What we do here



Source: https://en.wikipedia.org/wiki/Top_quark/

Data





$$(E, p_x, p_y, p_z)$$

Up to 200 particles per jet * 4 dimensions per particle = 800 entries per row

We work with 2 datasets:

- TopK - 2kk entries (used for most of our experiments)
- JetClass - 100kk entries (to get reference numbers)

Approaches

What we worked with

Models:

- Classic Deep Neural Network
- DeepJet CNN
- Particle Transformer (SOTA)

Losses:

- Cross-Entropy
- Focal

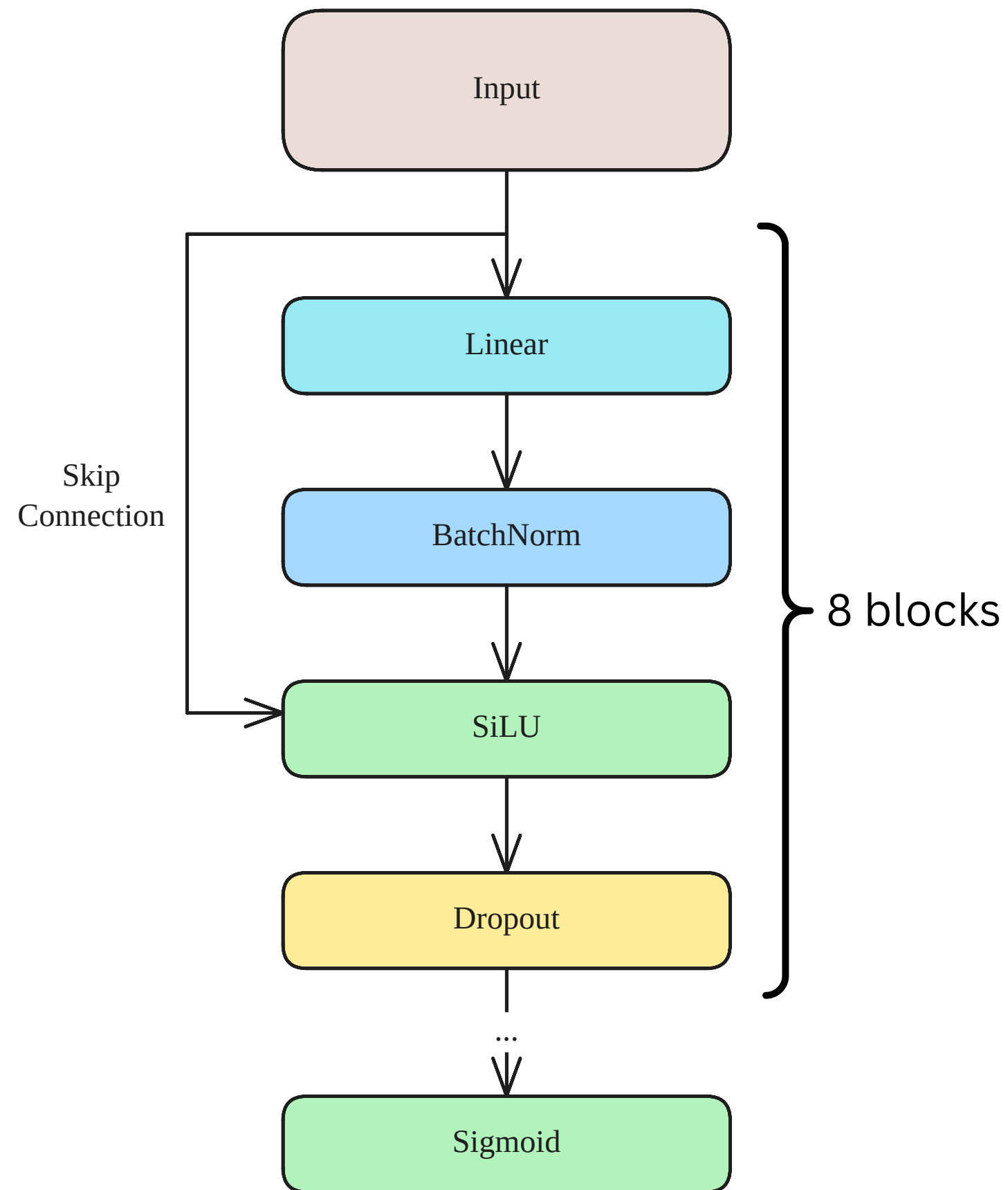
Optimizers/Schedulers:

To be discussed later

Metrics:

- Accuracy
- ROC/AUC

Base FFN

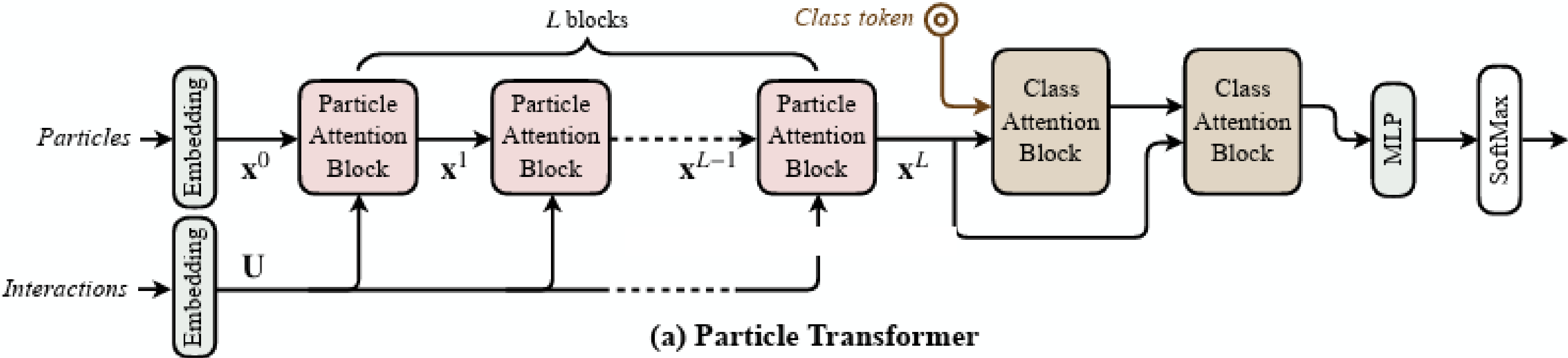


Optimizer: **AdamW**
Scheduler: **CosineAnnealing**
Number of Epochs: **50**

Initial Accuracy: **80%**
Final Accuracy: **89%**

ParT

Architecture



Source: <https://arxiv.org/abs/2202.03772>

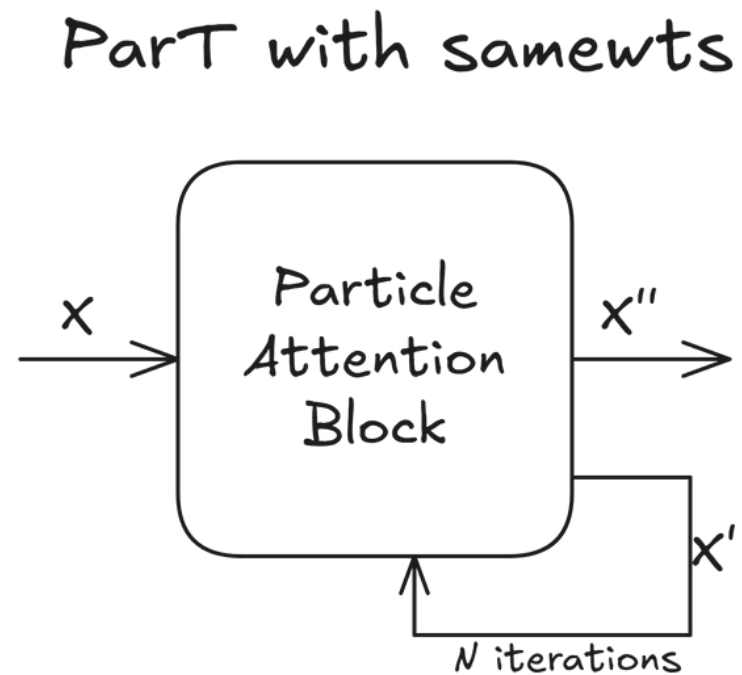
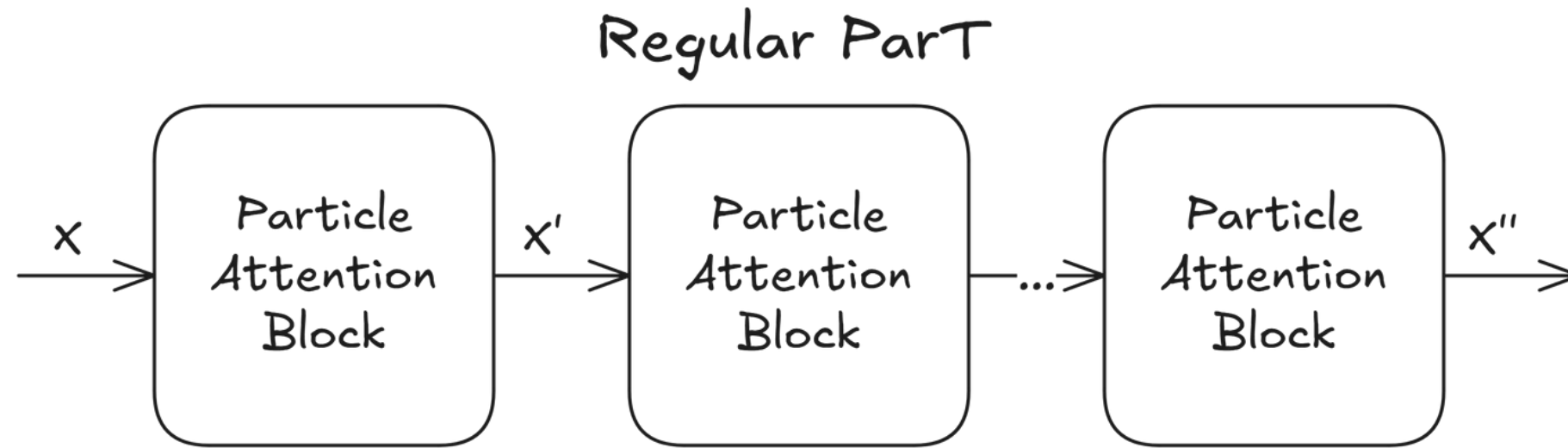
$$\text{Softmax}(QK^T + \text{mask})V$$

Self-attention equation with a mask

$\mathbf{U}$

Modifications

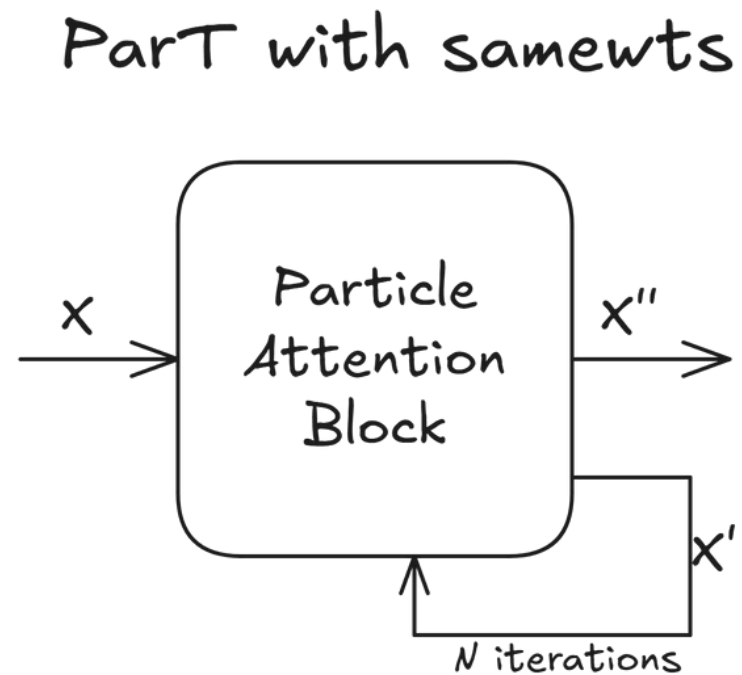
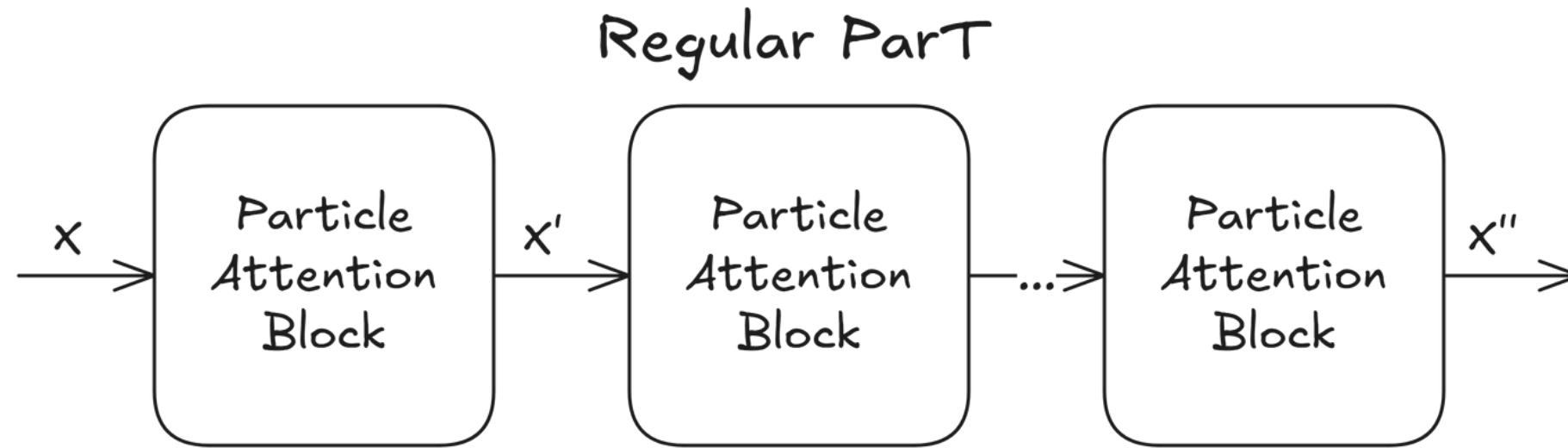
Same Weights



The weights are shared:

- between **Particle Attention blocks** (encoders)
- between **Class Attention Blocks** (decoders)

Same Weights



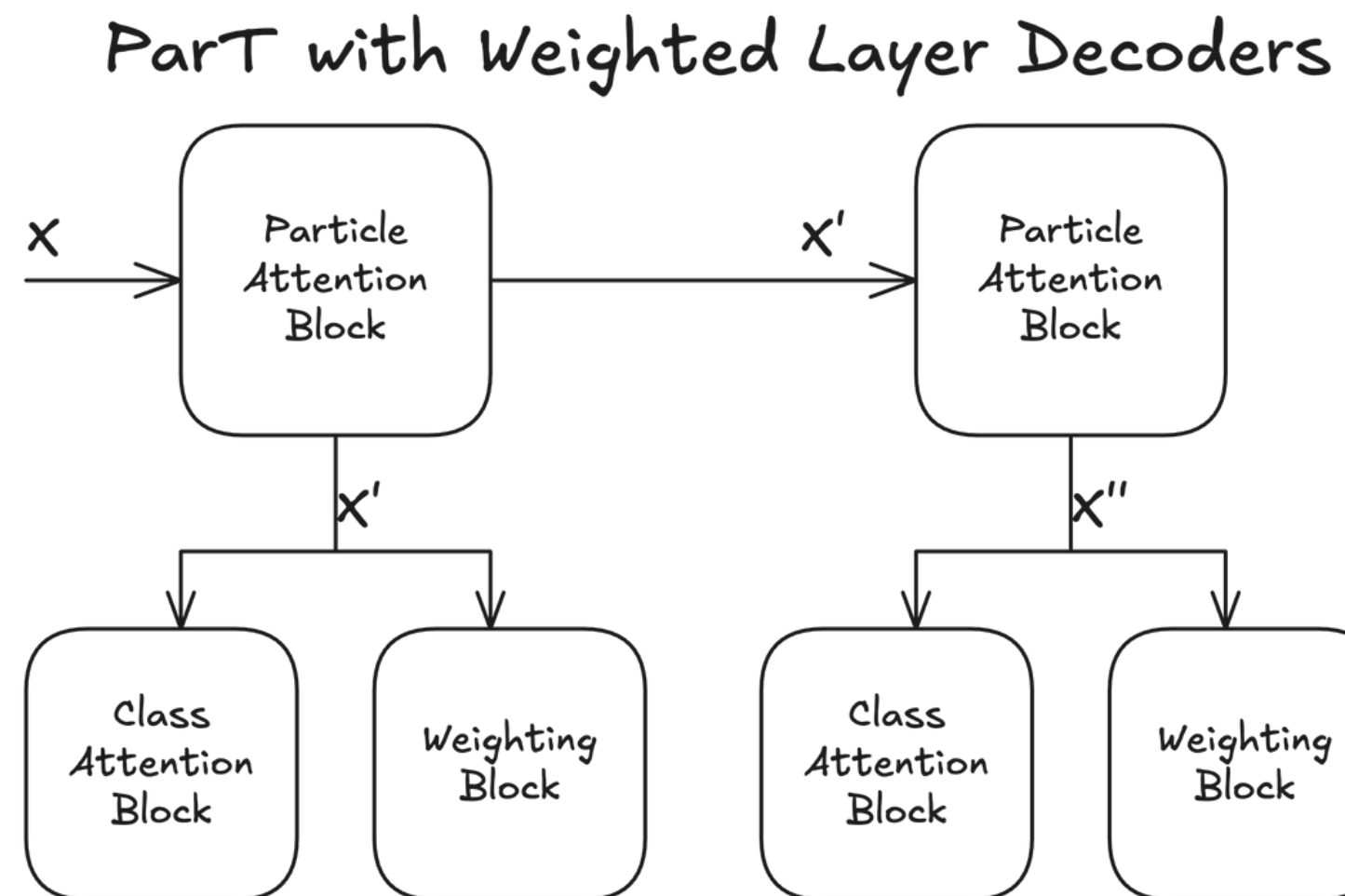
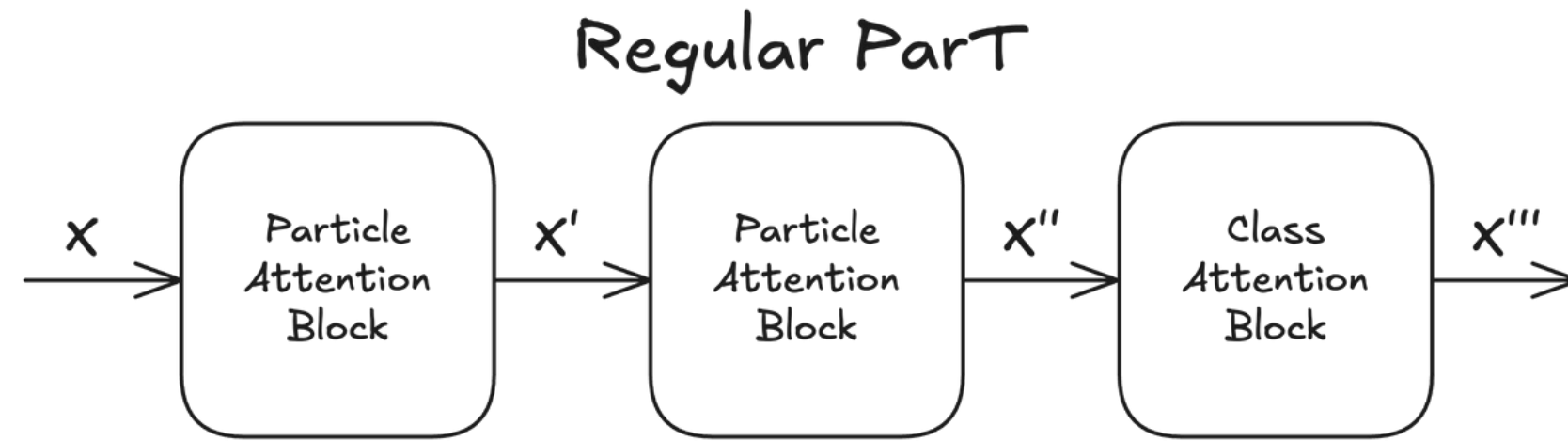
On **TopT**, this allowed to:

- increase accuracy to **94.2%** (**+0.2%**)
- decrease the number of weights by **75%**

On **JetClass**, this allowed to:

- increase accuracy to **86.19%** (**+0.1%**), while keeping the same number of parameters
- increase accuracy to **86.46%** (**+0.36%**), when scaling the model up

Weighted Layer Decoders

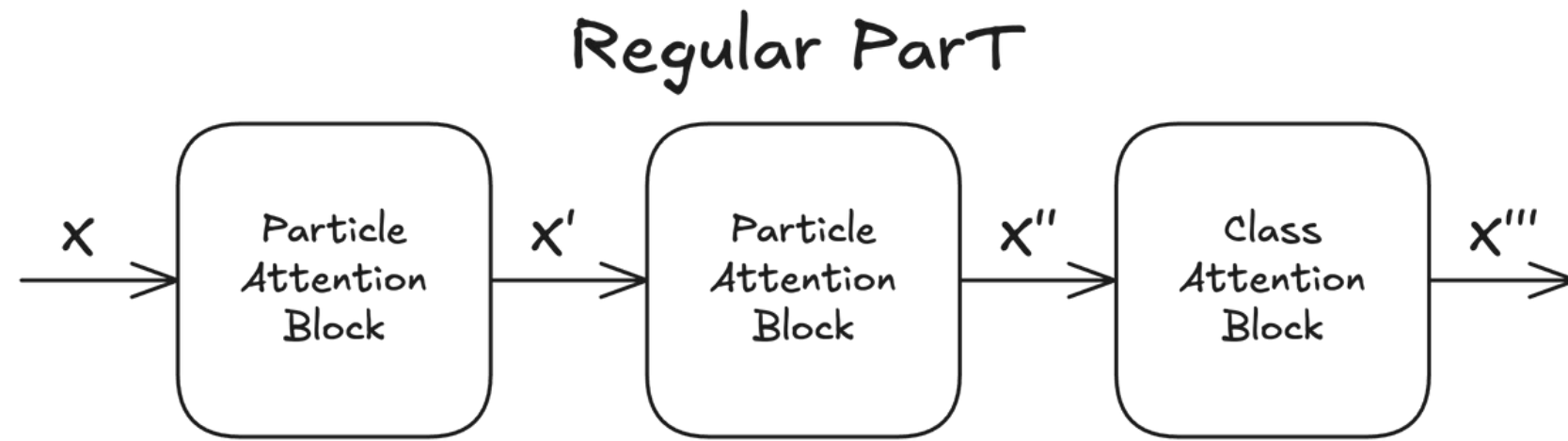


Each of the 28 sequential Particle Attention blocks is decoded by **2 decoders:**

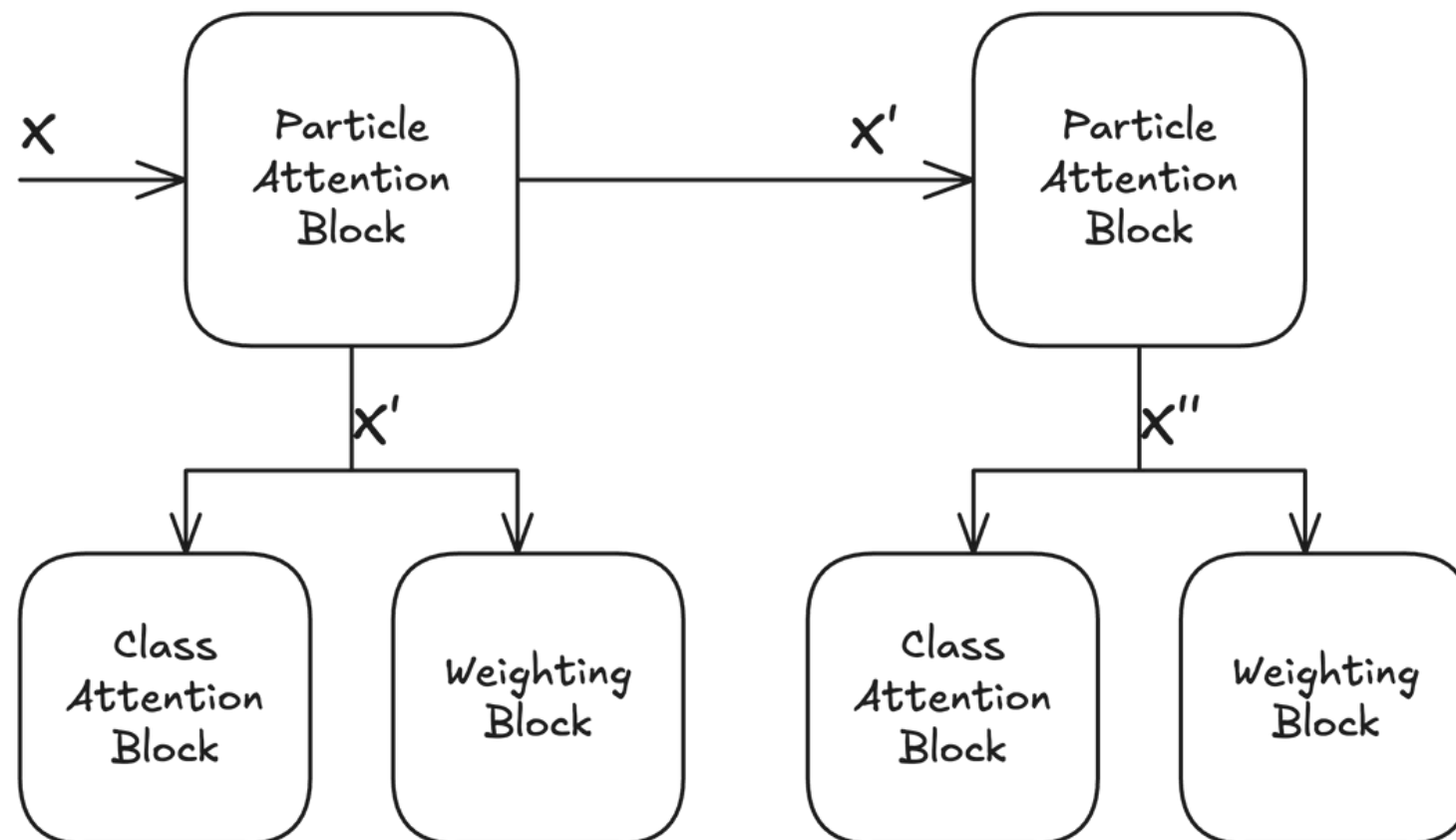
- a decoder that's used for **weights**
- a decoder that's used for **predictions.**

28 predictions are weighted to obtain the result

Weighted Layer Decoders



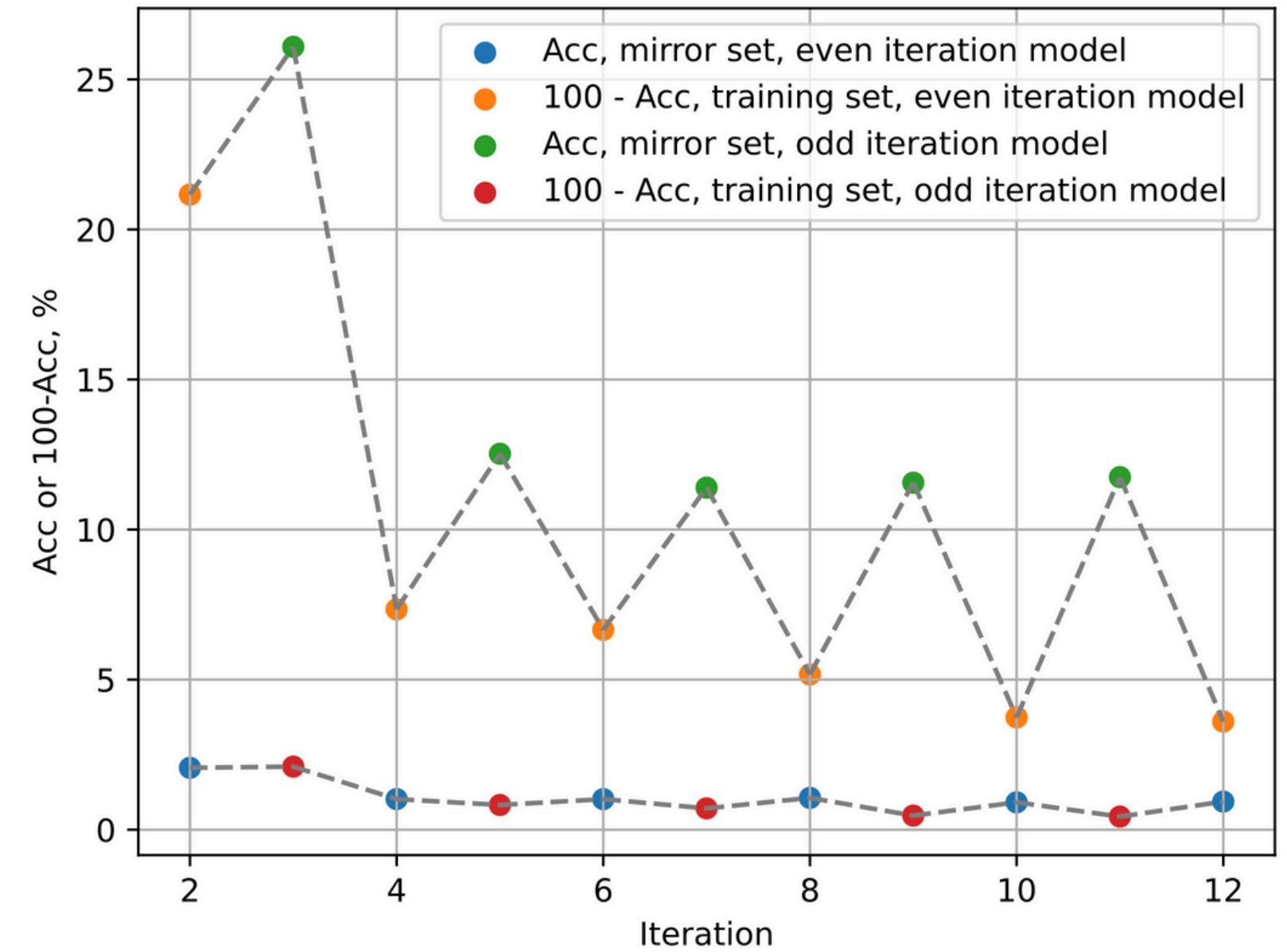
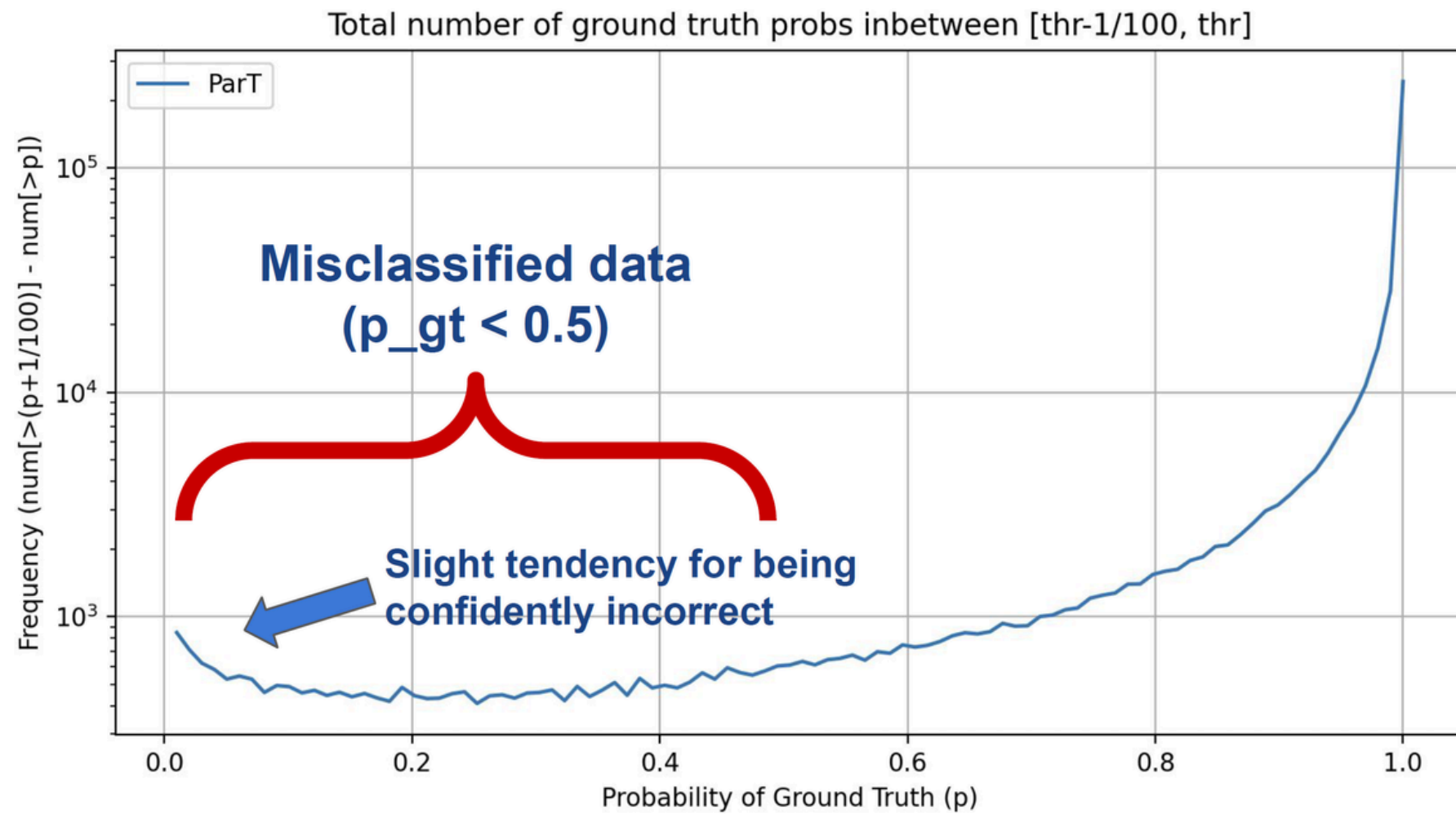
ParT with Weighted Layer Decoders



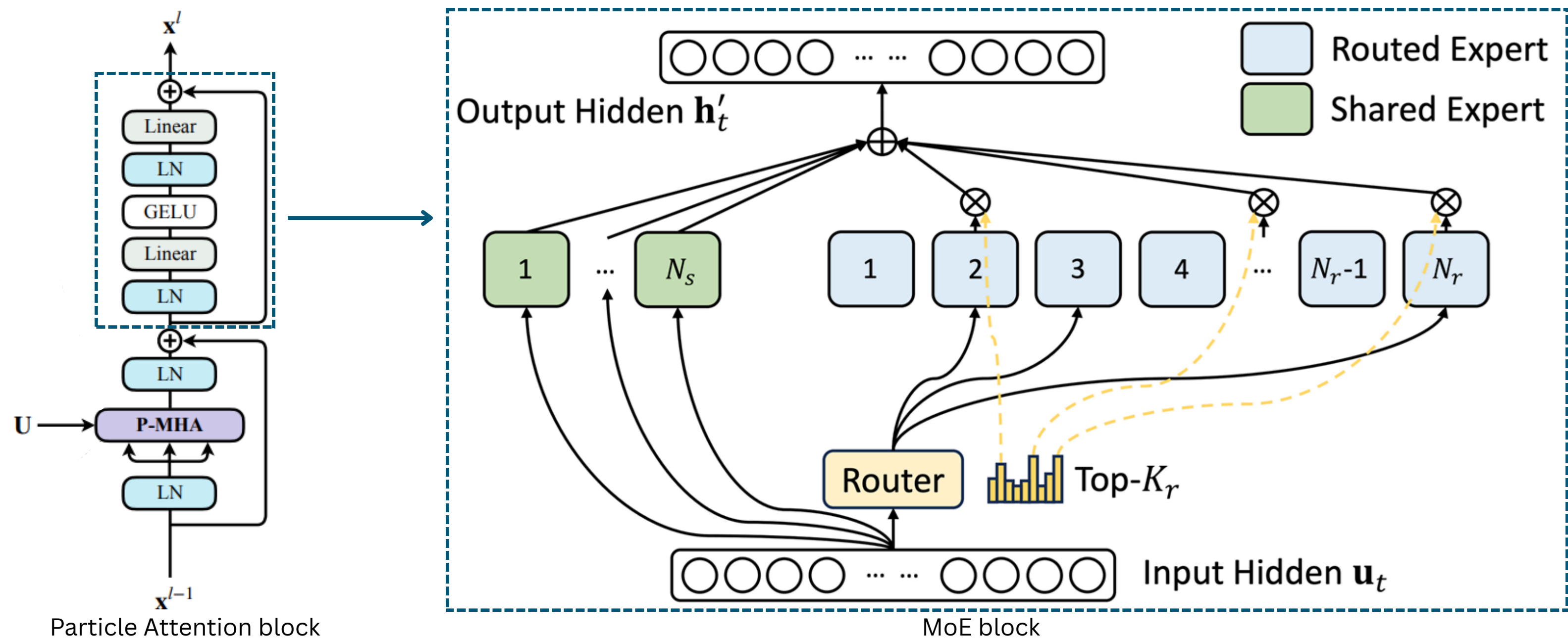
This allowed to:

- increase accuracy to **94.25%** on TopT (+0.25%)
- decrease the number of weights by **91%** (197K from 2.14M)

Mirrored Particles



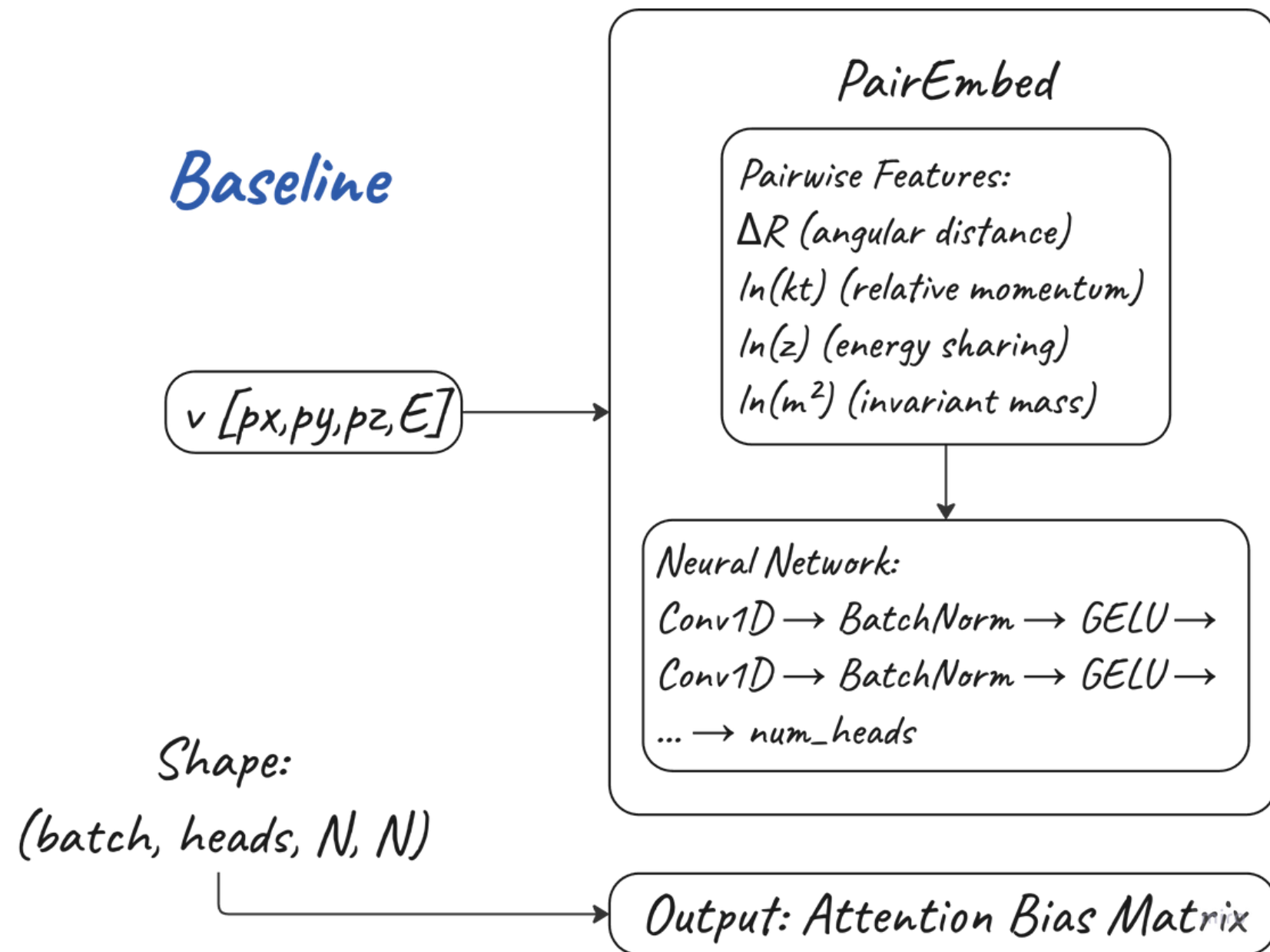
Mixture of Experts



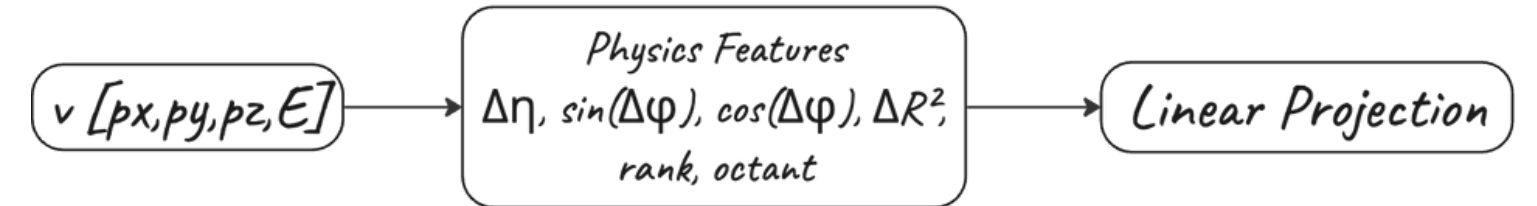
Attention Masks

$$\text{Softmax}(QK^T + \text{Mask})V$$

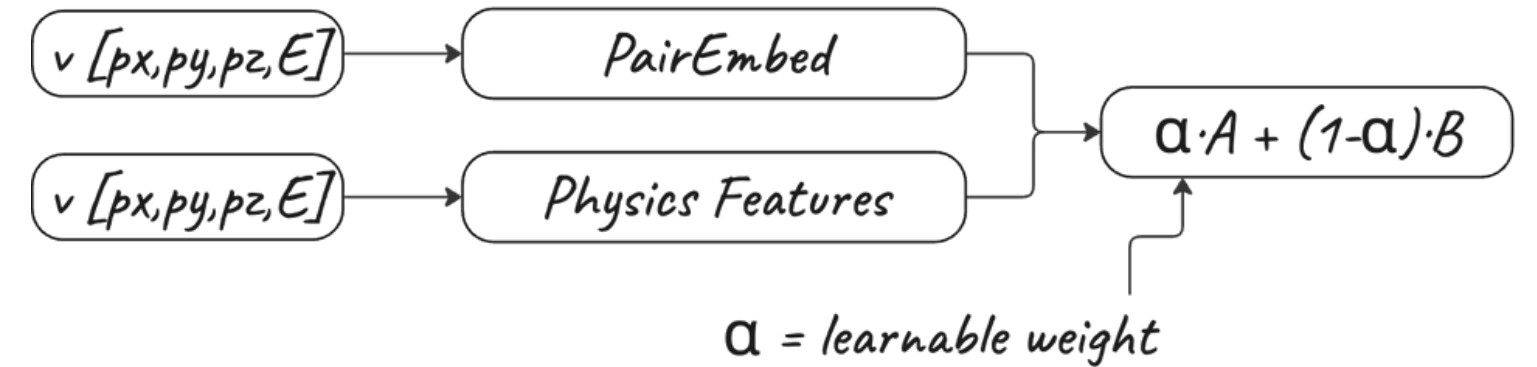
Self-attention equation with a mask



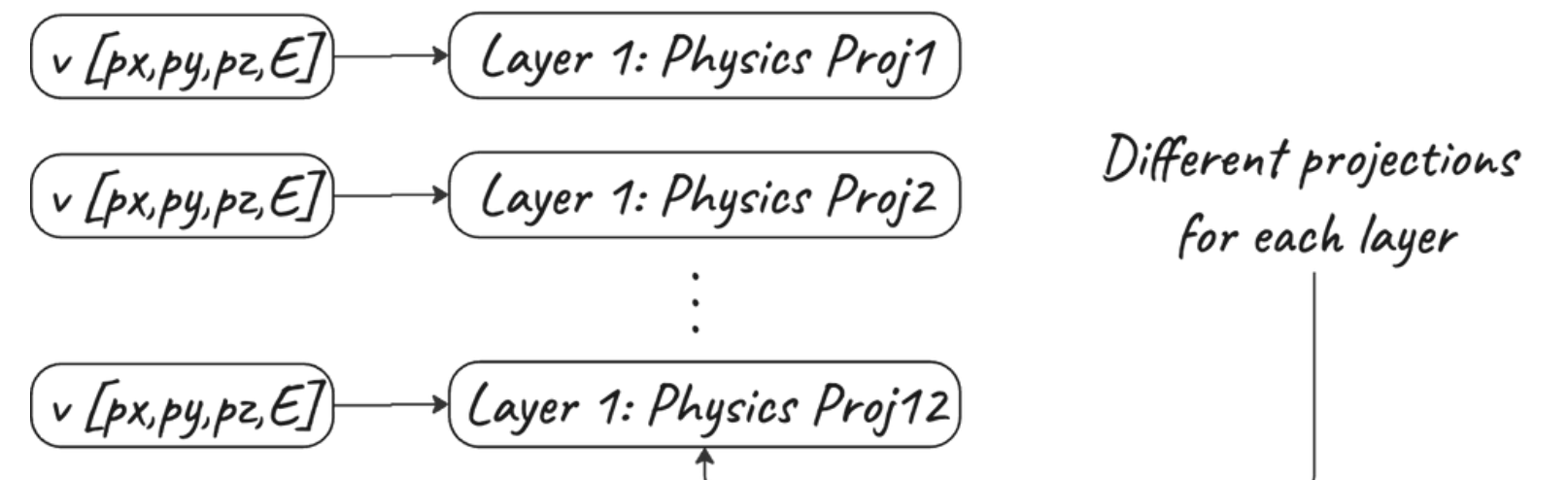
ModU (Hardcoded)



ModUHybrid (Fusion)

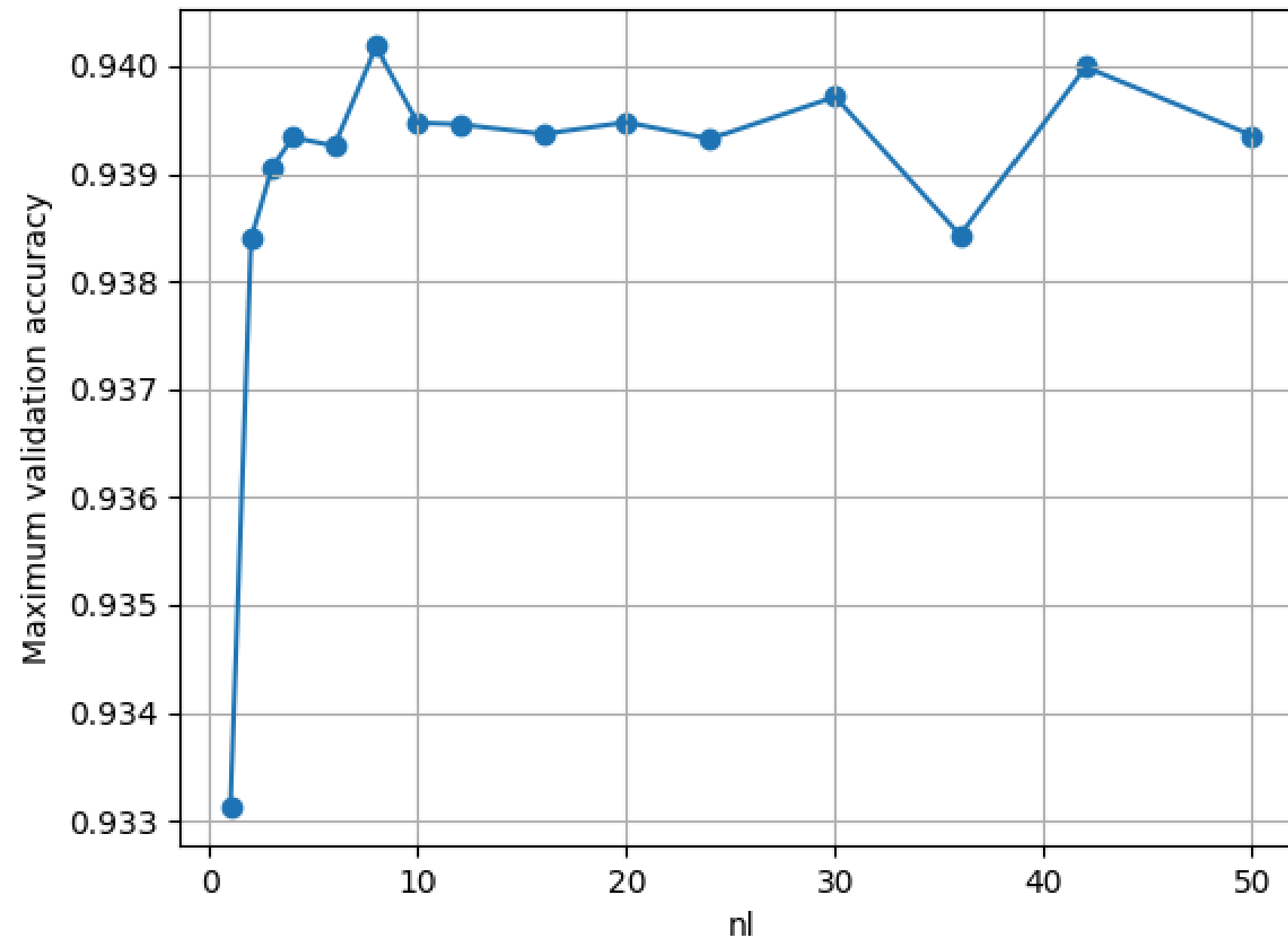


ModULayerwise (Layer-specific)

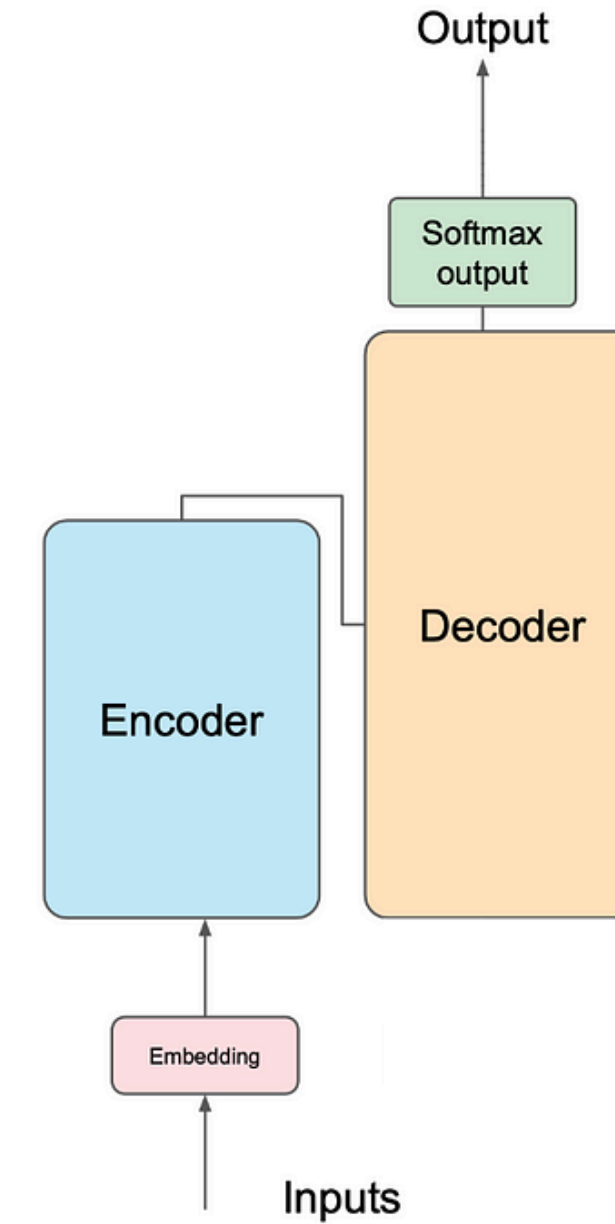


Experiments

Number of layers



Varying the total number of attention blocks (**nl**)



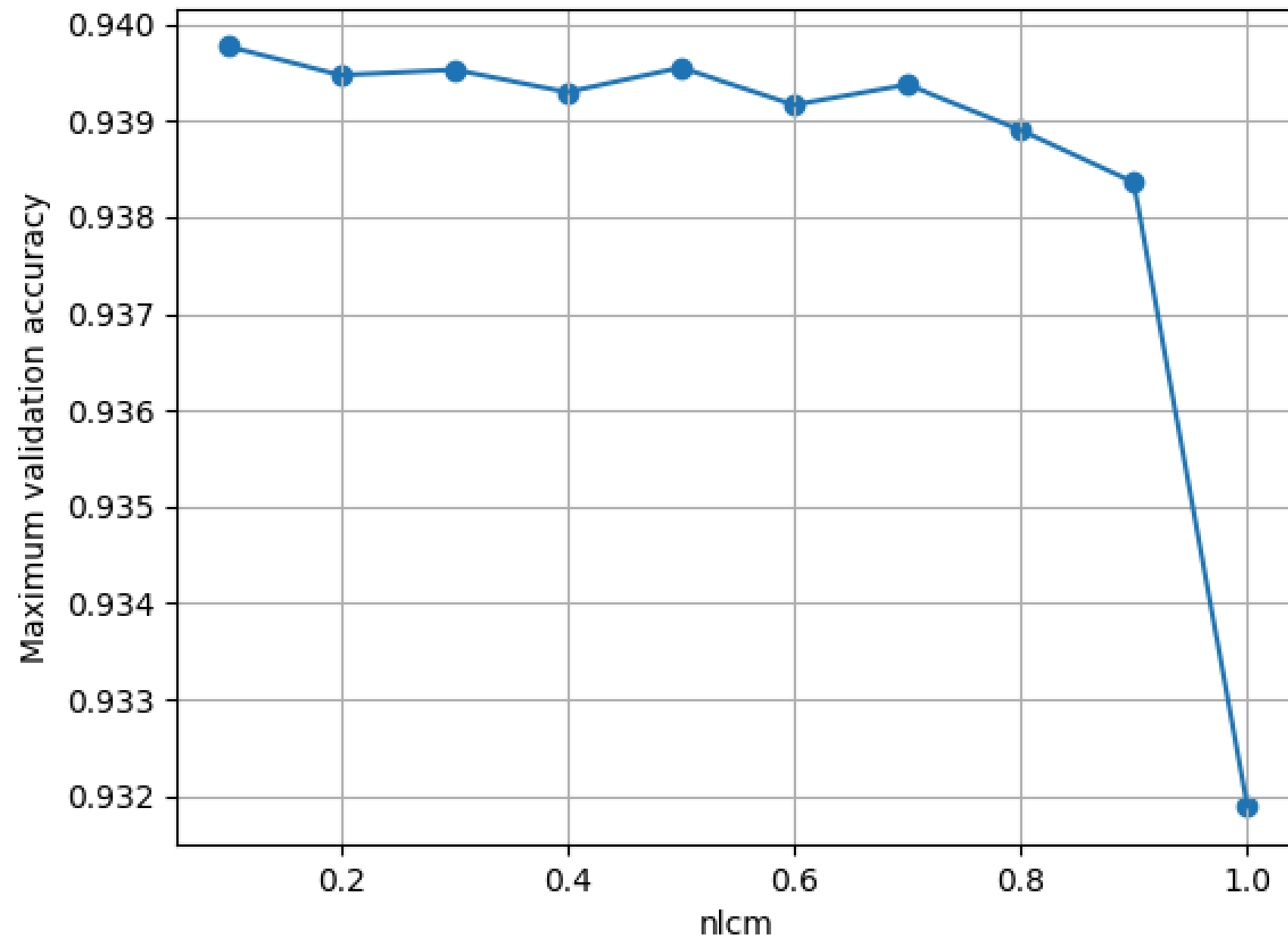
$$\mathbf{nl} = \mathbf{ne} + \mathbf{nd}$$

ne – number of encoders

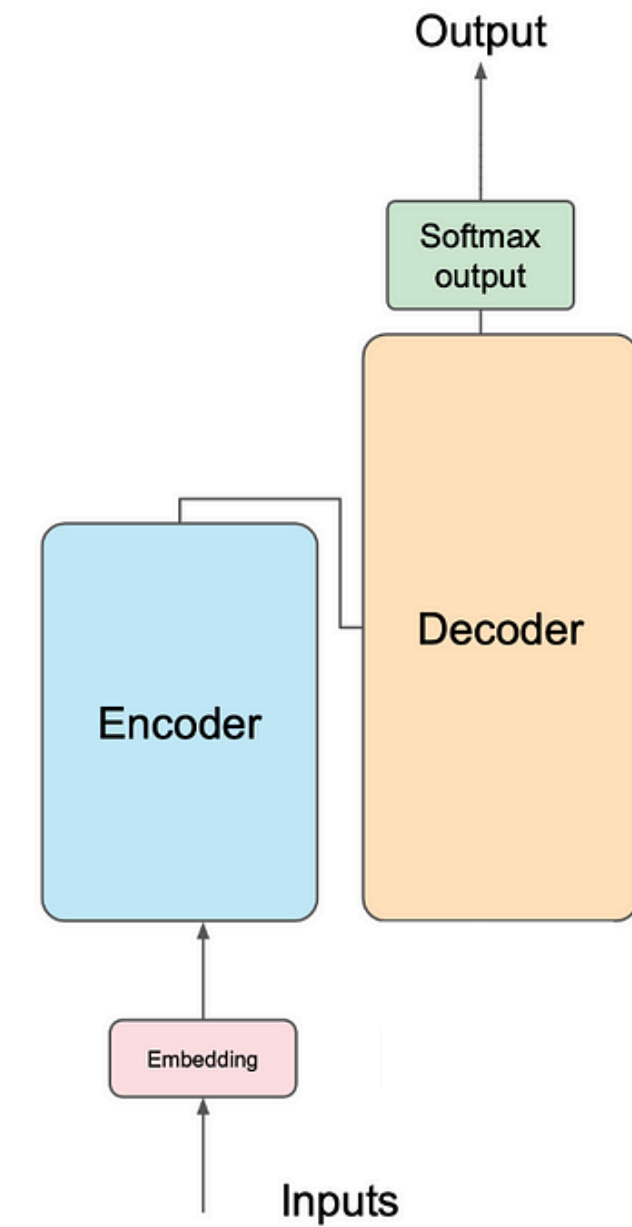
nd – number of decoders

*default: **nl** = 10*

Fraction of classification layers



Varying the fraction of Class-Attention decoders (**nlcm**)



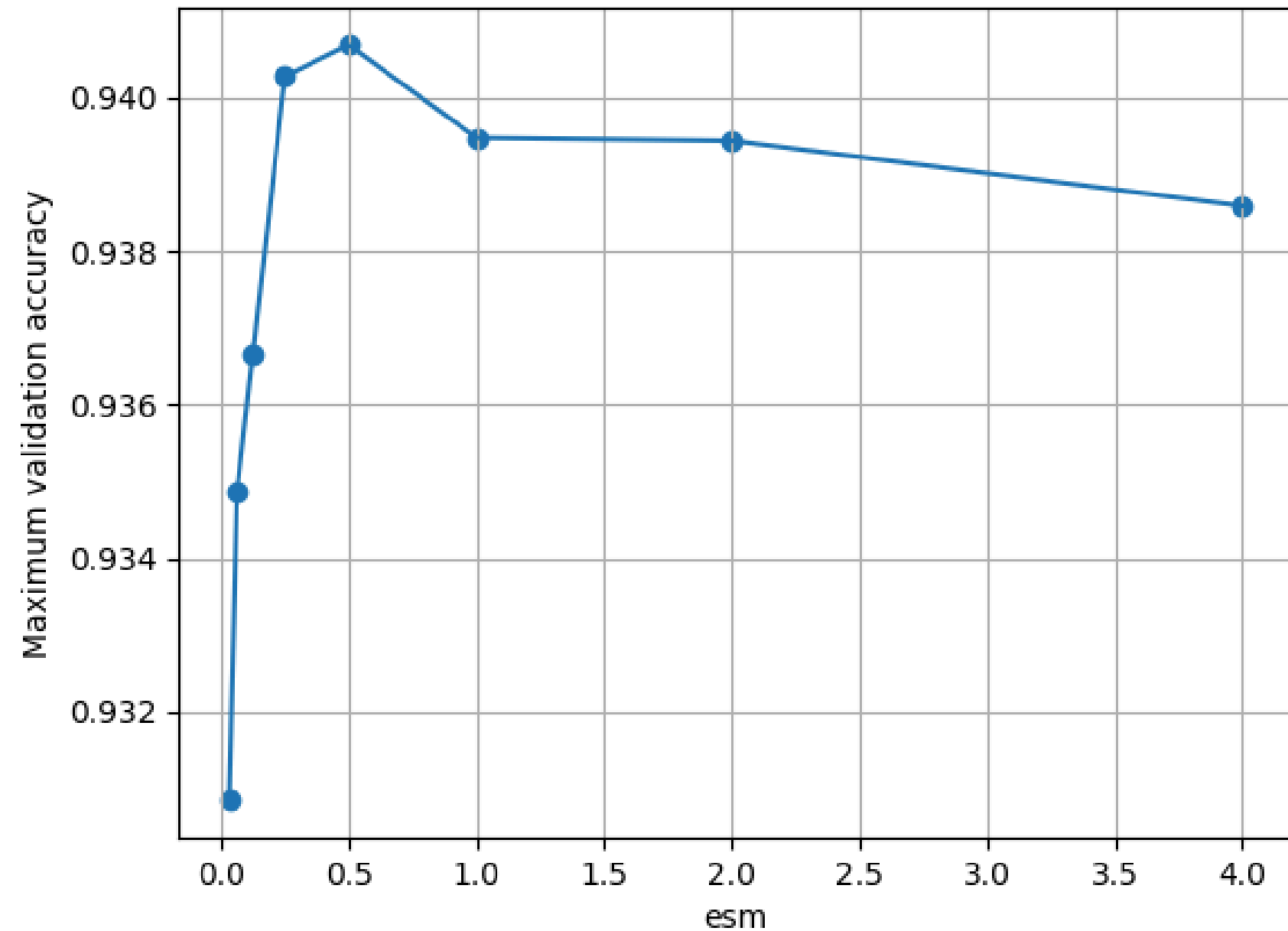
$$\text{nlcm} = \text{nd} / \text{nl}$$

nd – number of decoders

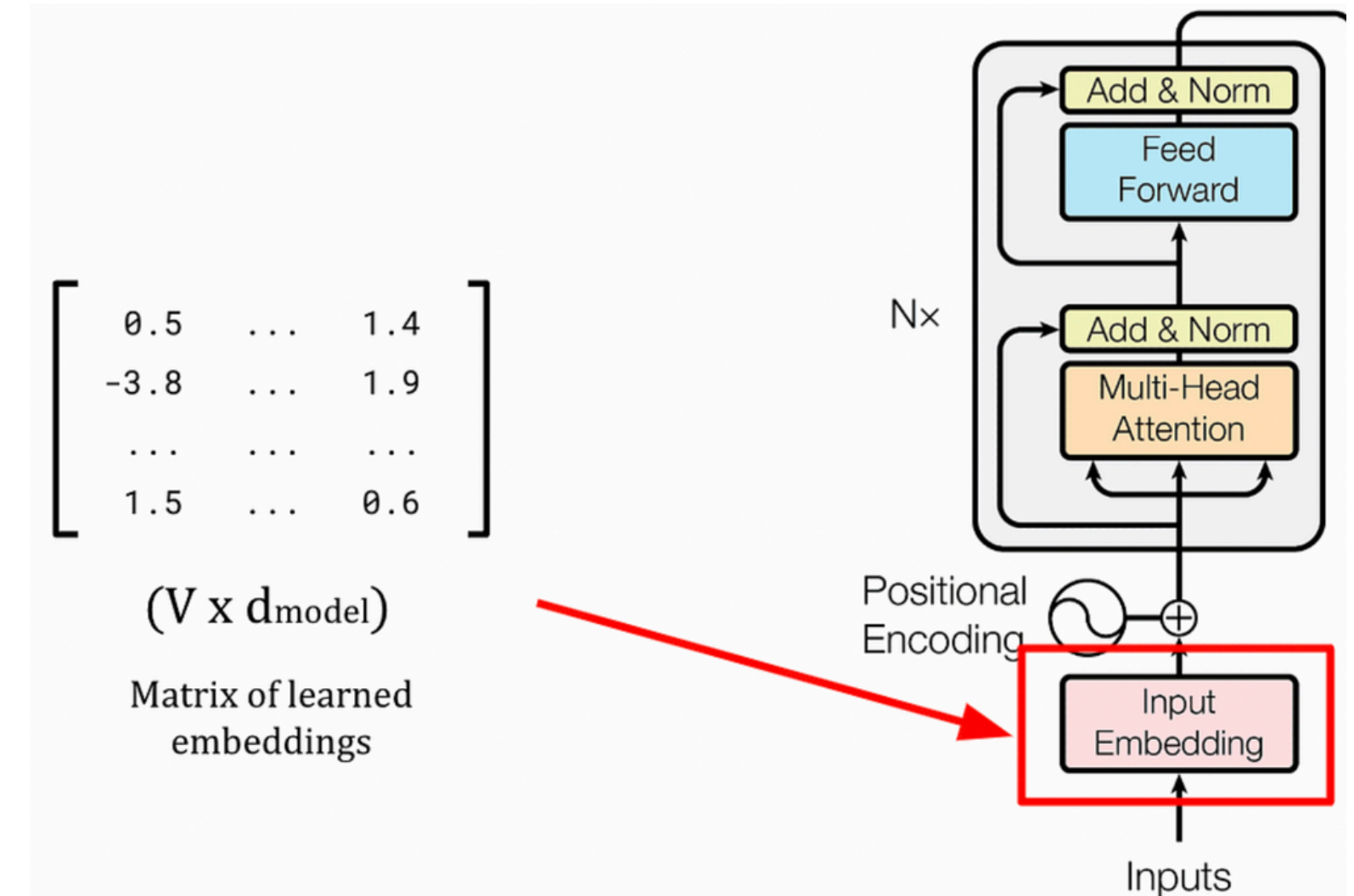
nl – total # of attn. blocks

*default: **nlcm** = 0.2*

Embedding dimension scaling



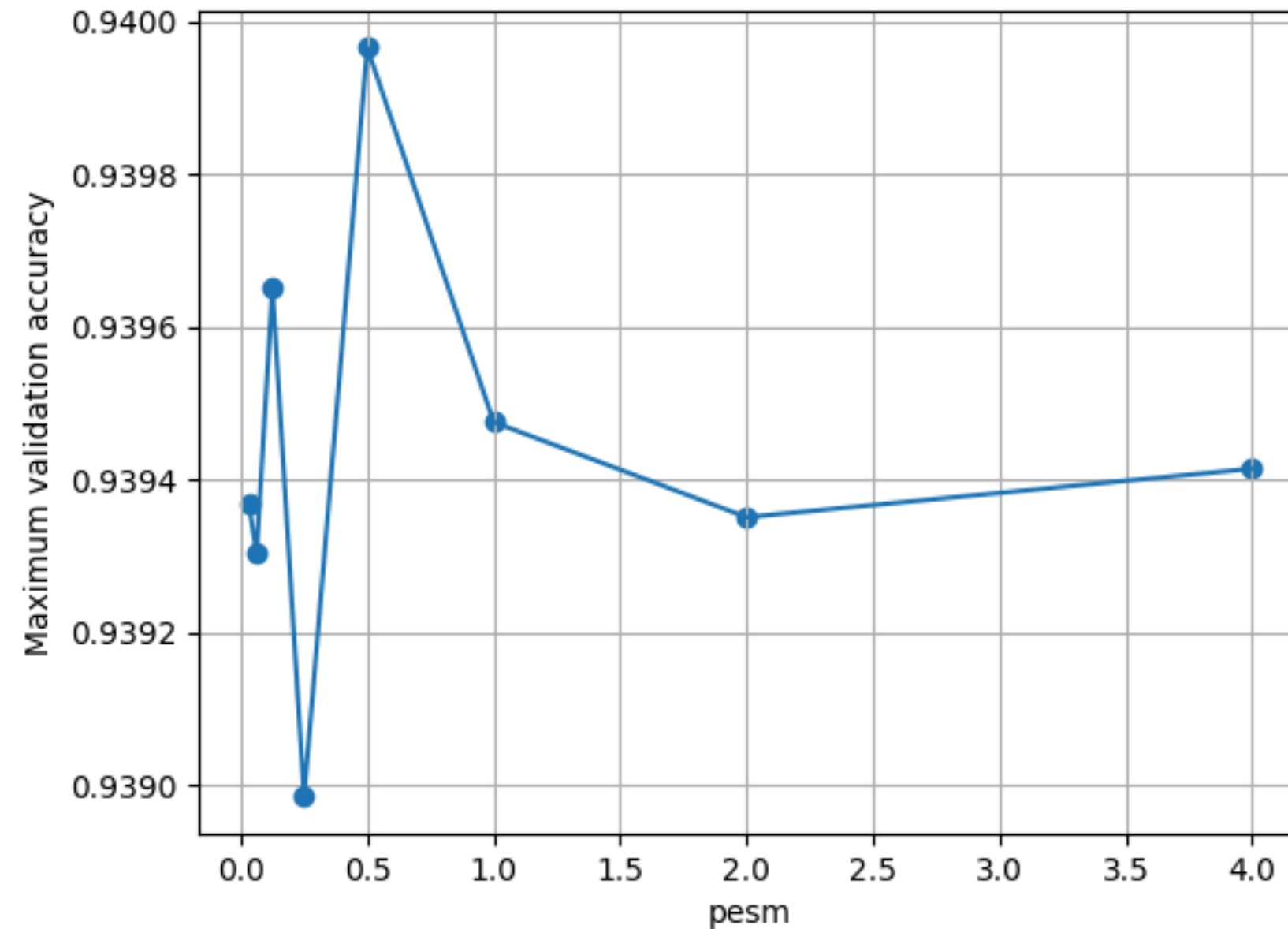
Scaling the embedding size (**esm**)



$$\mathbf{e} = \mathbf{esm} * 128$$

default embedding
size: **e = 128**

Particle Embedding dimension scaling

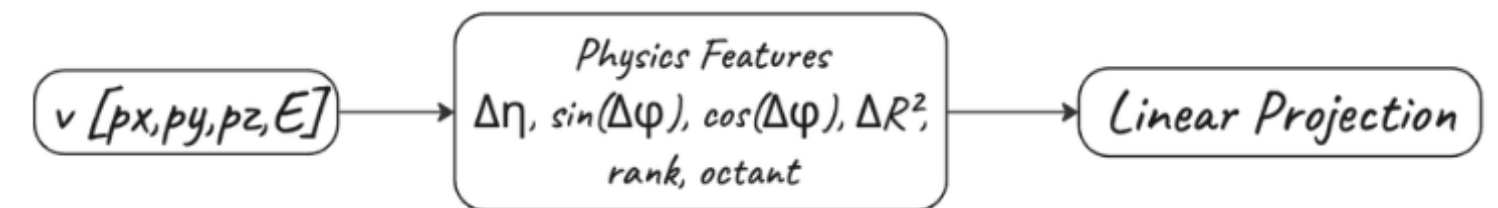


Varying the pair-embedding size (**pesm**)

$$\text{Softmax}(QK^T + \text{mask})V$$

Self-attention equation with a mask

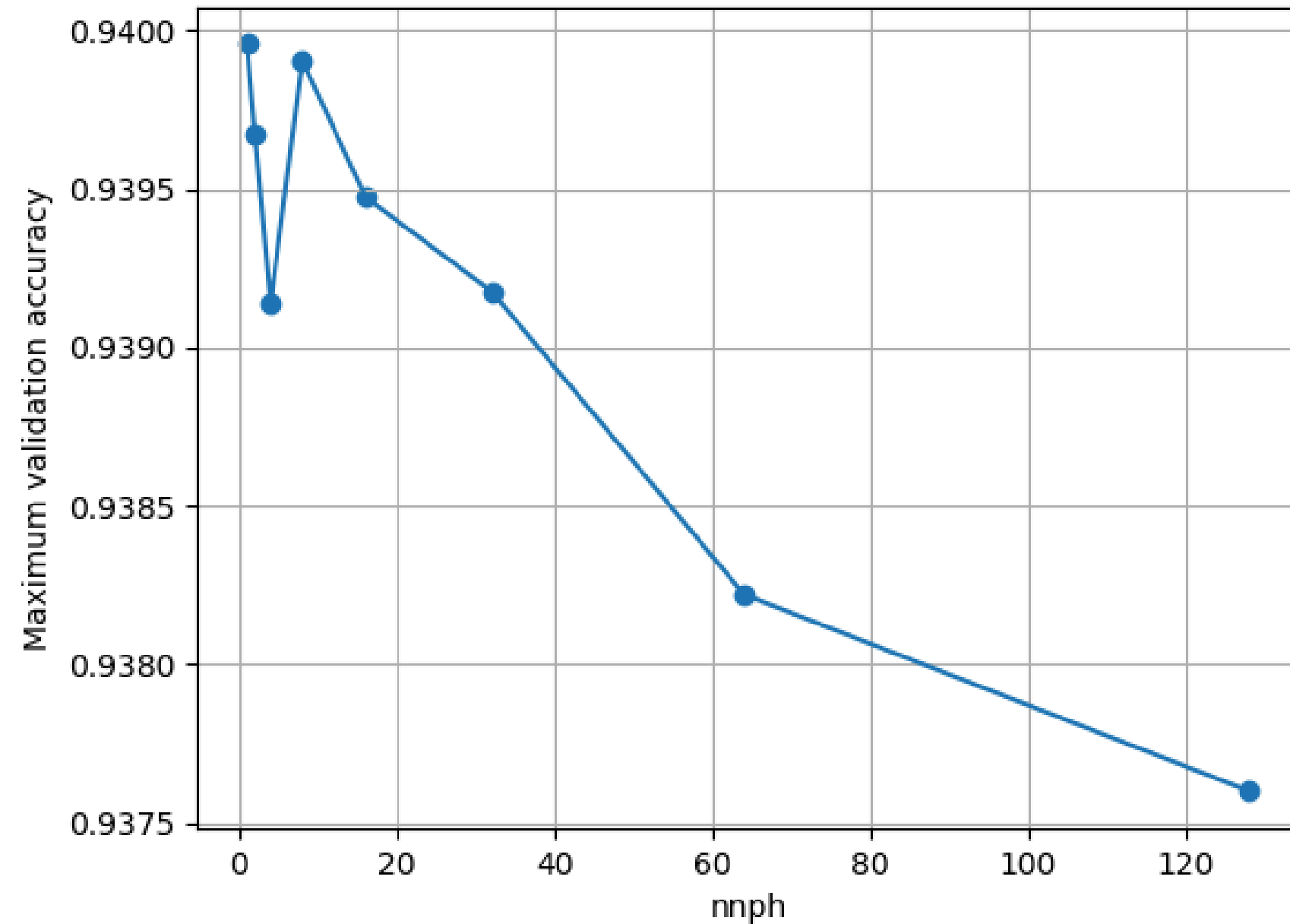
ModU (Hardcoded)



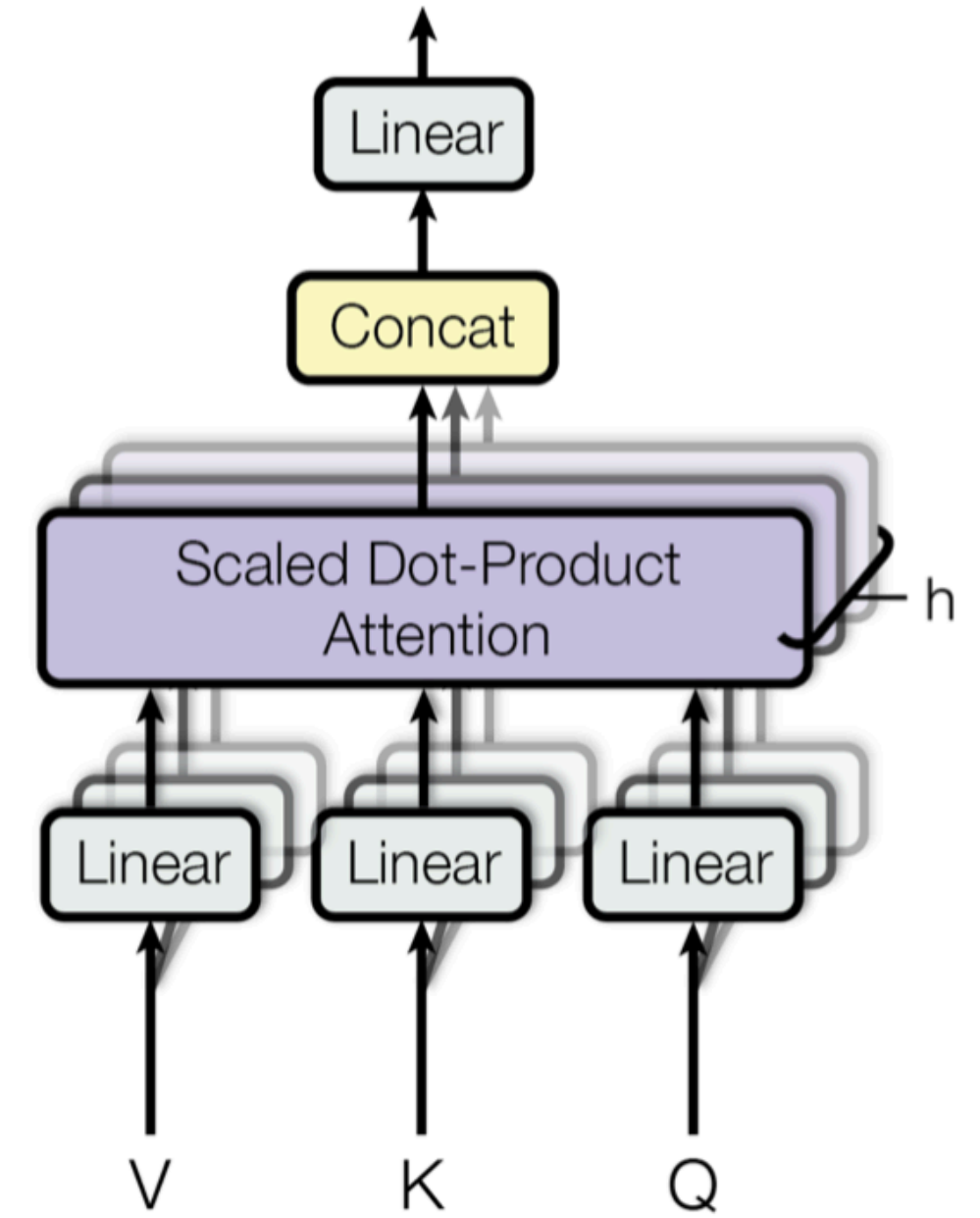
$$\mathbf{pe} = \mathbf{pesm} * 64$$

default pair embedding
size: **pe = 64**

Number of neurons per head



Varying the number of neurons per head (**nnph**)



$$\text{nnph} = e / h$$

e – embedding size

h – number of heads

Optimizers

1st-order:

- Ranger (default)
- AdamW
- Lion

2nd-order:

- Shampoo
- SOAP (best)

Results

Initial: 20 epochs, 93.8% accuracy

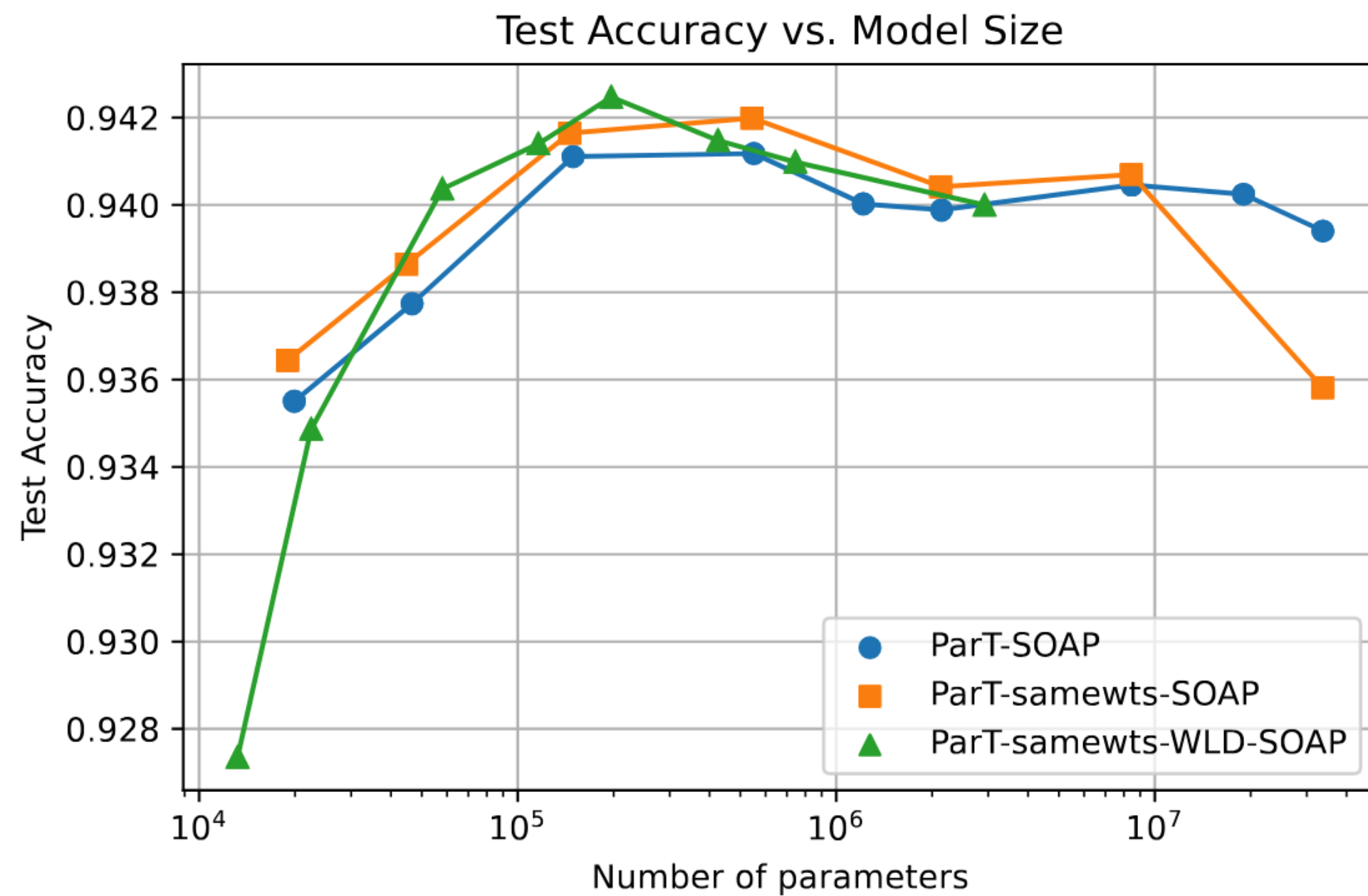
Final: 10 epochs, 94.05% accuracy

Failures

- Scalable Softmax
- torch-compile
- Transformers without Normalization
- Mirrored Particles

Results

Scaling experiments on TopK



Experiments on JetClass

Model Name	Test Acc.	#Params (M/k)	Comp. Complexity	Same-wts	ESM Val	Optimizer	Decay Frac.
ParT-samewts-ranger	85.69	545.98 k	306.18 MMac	✓	1	ranger	0.3
ParT-samewts-SOAP	85.71	545.98 k	306.18 MMac	✓	1	soap	0.3
ParT-esm05-SOAP-lrd07	85.62	552.01 k	133.65 MMac	✗	0.5	soap	0.7
ParT-samewts-esm2-ranger	86.08	2.13 M	991.55 MMac	✓	2	ranger	0.3
ParT-samewts-esm2-SOAP-lrd07	86.37	2.13 M	991.55 MMac	✓	2	soap	0.7
ParT-ranger	86.06	2.14 M	306.18 MMac	✗	1	ranger	0.3
ParT-SOAP	86.10	2.14 M	306.18 MMac	✗	1	soap	0.3
ParT-samewts-esm4-ranger	86.35	8.45 M	3.27 GMac	✓	4	ranger	0.3
ParT-samewts-esm4-nnph4-ranger	86.36	8.45 M	3.78 GMac	✓	4	ranger	0.3
ParT-esm2-SOAP-lrd07	84.90	8.47 M	991.22 MMac	✗	0.5	soap	0.7

Acknowledgements

- Pavlo Kashko - for the mentorship and support throughout the project
- CERN CMS team - for reviewing our work and suggesting directions for future research
- CERN CMS infra team - for allocating compute resources for our experiments
- The teaching team of Machine Learning, for making this project possible

QnA